Internship Finder: Design and Agent Prompt

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Product brief

A private, single-user web app that finds summer 2027 software internships you can actually take, ranks them against your profile, and drafts outreach where no role is open. It runs entirely on free tiers.

User: one person, no accounts, no auth.

Target: summer 2027 internships. Remote roles open to Nigeria (US, UK and worldwide employers) and onsite roles in Lagos.

Profile the matcher uses (from your resume):

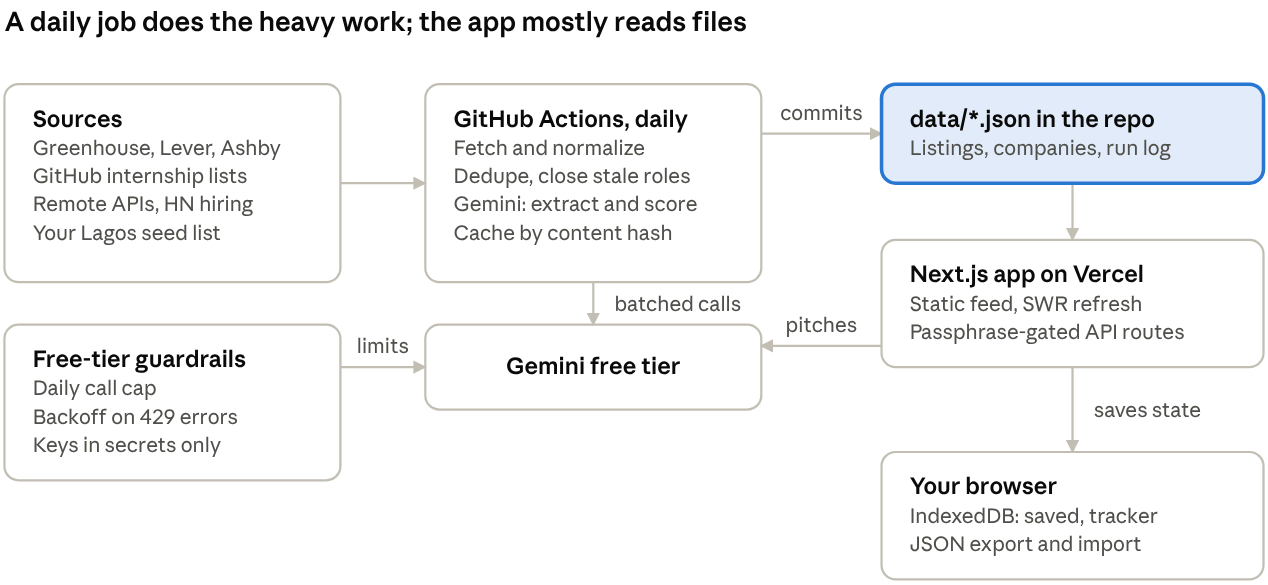
Field	Value
Stage	3rd year B.Sc. Software Engineering, Pan Atlantic University, Nigeria
Core skills	React, Next.js, JavaScript, HTML, CSS, Tailwind CSS
Backend and tools	Supabase, Git, GitHub, Vercel
Projects	Finance Manager PWA, Aurelia Grand Resort demo site, developer portfolio
Level	Beginner to intermediate; frontend first, open to other departments at beginner level
Availability	Summer 2027

Your resume summary is written for freelance work and says 10-15 hours a week. For internship applications, add a second summary line aimed at internships, and set full-time summer availability in the config file so the pitch drafts do not repeat the freelance wording.

Hard constraints: 100% free, Gemini free tier for AI, no scraping of sites that forbid it, and every listing links to its original source.

Architecture

A daily GitHub Actions job does the fetching and scoring, so the app only reads files and stays free.



architecture · 6 components, 1 daily job

The pipeline is the only scheduled spender of Gemini quota. The app reads the committed files, and only pitch drafts call Gemini live, through a passphrase-gated route. Your saved listings and tracker stay in your browser unless you add sync in phase 6.

Data sources

Use official public job-board APIs and community lists first; treat scraping as a last resort. The agent must test every source in phase 1 and drop any that fails, because endpoints and repo names change (names below are from memory, so approximate).

Source	Covers	Method	Reliability
Greenhouse, Lever, Ashby public job-board endpoints	Thousands of tech companies, US, UK and remote	JSON over HTTP, no login, one request per company slug	High

Source	Covers	Method	Reliability
Community internship lists on GitHub (Simplify and Pitt CSC style summer lists)	US and some remote summer 2027 roles, with apply links	Read the repo's listing file	High, US-heavy
Remote job APIs (Remotive, Arbeitnow and similar)	Remote roles worldwide, UK and EU leaning	Public JSON	Medium; few true internships
Hacker News "Who is hiring"	Startups, often remote	Public HN API, monthly thread	Medium
Your Lagos seed list	Fintechs, banks, startups, agencies in Lagos	A file you maintain by hand: company, careers URL, notes	High, because you verify it
Nigerian job boards (Jobberman, MyJobMag)	Onsite Lagos roles	Only if their terms and robots.txt allow it; otherwise a saved search link on your dashboard	Fragile
LinkedIn, Indeed, Glassdoor	Everything	Do not scrape	Off limits

Eligibility filter (required). Many remote US and UK internships need local work authorization. The AI step reads each description and returns one of: open to Nigeria or worldwide, region-locked, or unclear. Region-locked roles are hidden by default and shown on request.

Lagos gap. A lot of Lagos hiring happens on company career pages, X and LinkedIn posts. Expect to add companies to the seed list yourself; the app should make that a two-field form.

Freshness. Each listing stores the date it was first seen, last seen and, when the source gives it, the posted date. A listing not seen for 14 days is marked closed.

Features

Build order follows risk: listings and matching first, tracker next, contacts and pitches last.

Feature	What it does	Feasibility
Listing feed	Merges all sources, removes duplicates, shows duration, intern type, skills, deadline, location, apply link	Easy
Match score	Gemini scores each listing 0-100 against your profile and writes one "why this fits" line	Easy
Eligibility and scam flags	Marks region-locked roles; flags pay-to-apply, vague companies, Telegram-only contact	Medium
Application tracker	Saved, Applied, Interviewing, Offer, Rejected, with deadline countdowns	Easy
Company watchlist	Companies with no open role but a record of taking interns; each has a contact card	Medium
Contact finder	Suggests the right role to approach, pulls emails and names only if publicly listed on the company's own pages, and builds LinkedIn search links for you to click	Partial: verified emails are not reliable for free
Pitch drafter	Writes a short outreach email using your three projects and something specific from the company's own site	Easy
Weekly digest	New matches and closing deadlines sent by a free channel such as email or Telegram bot	Medium
Skill-gap view	Counts the skills most requested across saved listings that you do not list yet	Easy

Feature	What it does	Feasibility
CSV export and saved searches	Keeps your filters and lets you take data out	Easy

Pitch rules for the drafter: under 120 words, one specific line about the company, one project link, one clear ask (a 15-minute chat or who to send a CV to), and no flattery filler. It never invents facts about the company; if the site gave nothing specific, it says so and asks you for a line.

Contact rules: only publicly listed business contacts, never guessed or purchased email addresses, and every contact shows the page it came from.

Design system

Direction: luxury typography. Type does the work, color is nearly absent, and whitespace is generous. Think editorial magazine, not SaaS dashboard.

Token	Value
Display font	A high-contrast serif such as Cormorant Garamond, Fraunces or Playfair Display, loaded with next/font (free)
Body and UI font	A quiet sans such as Inter or Instrument Sans
Numerals	`font-variant-numeric: tabular-nums` on deadlines, durations, scores and counts
Headline	48-64px display on desktop, 32-40px on mobile, tight leading (1.05-1.15), `text-wrap: balance`
Body	16px, 1.6 line height, 62-70 characters per line
Labels	11-12px uppercase, 0.12em letter spacing, sans
Light mode	Background warm ivory `#F6F2EA`, text ink `#14110F`, hairlines `#D9D2C5`, muted text `#6B6258`
Dark mode	Background `#0F0E0D`, text `#EDE7DB`, hairlines `#2A2724`, muted text `#9A9184`

Token	Value
Accent	One only: brass `#B08A3E` (light) and `#C9A35B` (dark). Use it for "best match" and "closing soon" and nothing else. No purple, no gradients
Spacing	8px base grid; section gaps 64-96px; card padding 24px
Corner radius	One token, 12px for outer containers and 8px for anything nested inside. Use `corner-shape: squircle` where supported, with plain `border-radius` as fallback
Dividers	1px hairlines instead of boxes and shadows
Motion	150-250ms, ease-out, transform and opacity only; respect `prefers-reduced-motion`

Layout: a narrow reading column for the feed (max 720px), a wider two-pane view on desktop (list left, detail right), and a bottom sheet for detail on mobile.

Motion policy: animate only on a user action or a state change, never on a loop. Skeletons are static tinted blocks with no shimmer. The only moving loading indicator is a thin progress line that runs while a request is in flight and stops when it ends.

Contrast: body text must meet WCAG AA (4.5:1) in both themes; check the brass accent on the ivory background before using it for small text.

UI rules

Every tip, feature and law below is mapped to a place in this app. Where one does not fit a single-user tool, it says Skip and why, so the agent does not bolt it on.

The 20 UI tips

#	Tip	Where it applies	Use
1	Skeleton loaders	Feed cards and detail pane, same layout as the real content, static (no shimmer)	Yes

#	Tip	Where it applies	Use
2	Optimistic updates	Save, dismiss and status changes show instantly; roll back with a message if the write fails	Yes
3	Command menu	Cmd/Ctrl+K: jump to a listing or company, apply filters, toggle theme, add company	Yes
4	Shortcut hints	Key chips in the menu and on buttons: `/` search, `J` / `K` next and previous, `S` save, `D` dismiss	Yes
5	Undo toasts	After dismiss, delete or a status change; 6 seconds, one action	Yes
6	Stale-while-revalidate	Show the cached feed at once, refresh in the background, show "Updated 2h ago"	Yes
7	Pre-rendering	The pipeline writes JSON; Next.js statically generates the feed shell and listing pages	Yes
8	BlurHash placeholders	Only for photographic images such as the hero mockup; company logos do not need it	Conditional
9	View transitions	List to detail and filter changes, with a no-animation fallback	Yes
10	Swipe-to-dismiss sheets	Mobile listing detail and filters as bottom sheets (the vaul library)	Yes
11	Interruptible animations	Use transitions and springs that retarget mid-flight, not fixed keyframes	Yes
12	Predictive back	Push a history entry when a sheet opens so the back gesture closes it; full predictive back only matters for an installed Android PWA	Conditional
13	Hit slop	Icon buttons and chips get a 44x44px tap area through padding or a pseudo-element	Yes
14	Safe area insets	`env(safe-area-inset-*)` on sticky header, bottom sheets, tab bar; set `viewport-fit=cover`	Yes

#	Tip	Where it applies	Use
15	One-time code autofill	There is no login or OTP	Skip
16	Tabular numbers	Scores, deadlines, durations, counts	Yes
17	Balanced headlines	`text-wrap: balance` on all headings	Yes
18	Continuous corners	Squircle corners where supported, with a rounded fallback	Yes
19	Idempotency keys	Backend only: upsert listings by a hash of source plus external ID, cache Gemini results by listing hash, key each weekly digest by week	Yes
20	Amazon safe triangle	Only if a desktop filter menu has nested flyouts; Radix submenus already include it	Conditional

The 20 site features

Feature	How it is used here	Use
Sticky header	Wordmark, search field, theme toggle; the tracker's column headings stick too	Yes
Skip to content	First focusable element on every page	Yes
Password visibility toggle	Used on the passphrase gate that protects the Gemini-backed API routes, so nobody else can spend your free quota	Yes
UTM tracking	Add `utm_source=outreach&utm_campaign=<company>` to your portfolio link inside pitch drafts so you can see which pitches get clicks. Do not alter employers' apply links	Yes
Form success state	Add company, log application, save contact: inline check and short message, then reset	Yes
Loading animations	One thin progress line while a request runs, stopped when it ends; no looping spinners	Yes
Form error state	Inline, next to the field, linked with `aria-describedby`, says how to fix it	Yes

Feature	How it is used here	Use
Hover states	Rows, buttons and links respond on hover and on `:focus-visible`; transitions only	Yes
Scroll progress bar	2px line under the header on the long listing detail view	Yes
Copy button	Pitch text, email address, apply link; label changes to "Copied" for 2 seconds	Yes
Print stylesheet	Prints a listing or a pitch cleanly: no header, no buttons, black on white	Yes
Dark mode toggle	Follows the system by default, toggle overrides, saved, no flash on load	Yes
Simple cookie banner	Not needed while the app stores only a theme preference. Add one only if you add analytics	Skip
Site search	Client-side search over title, company, skills and notes (MiniSearch or Fuse), also reachable from the command menu	Yes
Back-to-top button	Appears after about 800px of scrolling on the feed	Yes
Mobile menus	Bottom tab bar (Feed, Tracker, Companies, Settings) plus a sheet for filters	Yes
Confirmation modals	Only for irreversible actions such as deleting a company or clearing all data. Reversible actions use an undo toast instead	Yes
Last updated date	"Data updated 10 Nov 2026, 06:00" on the feed; "last seen" on each listing	Yes
Expandable FAQ	Small accordion on the settings page: how scoring works, what flags mean, where data comes from	Yes
Floating contact	A single-user app has no one to contact. Repurpose it as a floating "Add company" button on mobile	Adapted

The 17 laws

Law	Applied as
Aesthetic-usability	The luxury typography is the feature: serif display, hairlines, generous whitespace, one accent
Hick's law	At most 5 visible filters (region, type, match, status, sort); the rest sit behind "More filters"
Fitts's law	Large tap areas; the primary action (Save) is the biggest target on a card; mobile actions sit in the thumb zone
Jakob's law	Familiar patterns: search at top, list left and detail right, bottom tab bar on mobile, Cmd+K
Miller's law	Cards show at most 5 facts; lists are grouped into chunks (This week, This month, Later)
Doherty threshold	Every interaction responds in under 400ms through cached data, skeletons and optimistic updates
Proximity	Duration, skills, deadline and location sit together in one block per card
Similarity	All "deadline" items share one style, all "skill" items share another, everywhere
Common region	One region per listing and one per tracker column, defined by a hairline, not nested boxes
Visual hierarchy	Company and role title first, then match and deadline, then details; the headline serif is the largest thing on screen
Progressive disclosure	Cards show essentials; the detail pane holds the full description, contacts and pitch
Recognition over recall	Saved searches, visible active filters, recent companies in the command menu
Von Restorff	The brass accent marks only "best match" and "closing soon", so they stand out

Law	Applied as
Peak-end	Make saving, applying and sending a pitch feel good (a clean confirmation); end each session with "You have 3 follow-ups this week"
Goal-gradient	Tracker header shows progress toward a goal you set, for example "8 of 15 applications sent"
Serial position	Best matches come first in the feed; in any row of actions, the two most used (Save and Apply) sit first and last, and rarely used ones sit in the middle
Tesler's law	Complexity is moved to the pipeline, not the user: the app does the eligibility checks, scoring and deduping so you only decide

Cleanup rules (apply to every screen)

- Remove purple gradients and any gradient used as decoration.
- Use real icons from one library only; no emoji as icons.
- Delete badges that carry no decision. Keep a badge only for match, closing soon, or region-locked.
- Flatten nested cards: one container level, separated by hairlines and spacing.
- Group related items under one label (dates together, skills together, contacts together).
- Use one corner radius token everywhere (12px outer, 8px nested).
- Stop looping animations. Animate only on user action or state change.
- Run a final pass: if an element does not help someone decide or act, remove it.

Components, icons and mockups

Components: shadcn/ui on Radix primitives with Tailwind. It matches the stack on your resume, the code lives in your repo so you can restyle it fully, and it ships the pieces this app needs. Add `cmdk` (command menu), `vaul` (swipe-to-dismiss sheets), `sonner` (undo toasts) and `motion` (interruptible animations).

Option	Verdict	Reason
shadcn/ui	Use	Unstyled enough to carry the luxury look; Tailwind native
Radix UI / Base UI	Use underneath	Accessible primitives (dialog, dropdown, tabs, accordion, submenu safe triangle)
HeroUI	Skip	Strong default look that you would have to fight
Material UI	Skip	Heavy, and its Material look clashes with editorial typography

Icons: Lucide React (the shadcn default), stroke width 1.5, one size scale (16, 20, 24). Phosphor React is a fine thin-weight alternative; pick one library and do not mix. No emoji as icons.

Mockup images. Use sparingly, for one hero visual and empty states, never as decoration on every card. Check each site's license terms before use, and add alt text. These are the sites you listed:

- Lummi
- Mockups Supply
- Moyo Studio

Generate a BlurHash for each image at build time so it fades in over a soft placeholder.

The agent prompt

Paste the prompt below into your agent, then paste the Design system, UI rules and Build phases sections of this doc beneath it so the agent has the full spec.

```
# Role
You are a senior full-stack engineer building "Internship Finder", a private
single-user web app. Work in the phases given below. After each phase, stop
and report: what works, how you verified it, what failed, and what you
changed from the plan.

# Goal
Find software internships for summer 2027 that I can actually take, rank them
against my profile, track my applications, and for companies with no open
role but a record of taking interns, give me a contact route and a short
```

```
pitch draft.

# My profile (store in config/profile.json)
- 3rd-year B.Sc. Software Engineering student, Pan Atlantic University,
Nigeria. Based in Lagos.
- Skills: React, Next.js, JavaScript, HTML, CSS, Tailwind CSS, Supabase, Git,
GitHub, Vercel.
- Projects: Finance Manager PWA (React/Next.js, Supabase, offline-capable),
Aurelia Grand Resort (responsive hospitality demo site, Tailwind), developer
portfolio (Vercel). Add live and repo URLs to the config.
- Level: beginner to intermediate. Frontend first. Also open to other tech
departments at beginner level.
- Target: summer 2027 internship, full-time availability for the summer.
- Regions: remote roles open to someone in Nigeria (US, UK, worldwide
employers), and onsite or hybrid roles in Lagos.

# Hard rules
1. 100% free. Use only free tiers: GitHub Actions, Vercel hobby, Gemini free
tier. No paid APIs, no credit card. Supabase is optional and only in the last
phase.
2. Never scrape LinkedIn, Indeed or Glassdoor. For any other site, read
robots.txt and the terms first. If scraping is not allowed, skip it and add a
saved-search link to the dashboard instead.
3. Never invent data. Every listing and every contact stores its source URL.
If a field is not in the source, store null.
4. Never guess, generate or buy email addresses. Contacts come only from the
company's own public pages. Otherwise suggest the role to approach and give a
LinkedIn search link for me to click.
5. The Gemini API key lives only in GitHub Actions secrets and Vercel server
environment variables. It never reaches client code.
6. Respect free limits: cache AI results by content hash, batch listings per
call, retry 429 errors with exponential backoff, and stop at a daily call cap
set in an environment variable. Read the model name from GEMINI_MODEL (check
Google AI Studio for the current free model).
7. Protect all API routes with a passphrase gate (APP_PASSPHRASE) so nobody
else can spend my quota.

# Stack
Next.js App Router, TypeScript, Tailwind, shadcn/ui on Radix, lucide-react,
cmdk, vaul, sonner, motion, TanStack Query (stale-while-revalidate),
MiniSearch. The pipeline is a Node script (scripts/pipeline.ts) run daily by
a GitHub Actions cron job that commits data/listings.json and
data/companies.json. Personal state (saved, dismissed, tracker, watchlist,
```

```
notes) lives in IndexedDB with JSON export and import.

# Data model
Listing: id, source, external_id, content_hash, company, title, location,
work_mode (remote, onsite, hybrid), eligibility (open_to_nigeria,
region_locked, unclear), duration, intern_type, skills[], requirements,
deadline, start_window, apply_url, source_url, first_seen, last_seen, status
(open, upcoming, closed), match_score (0-100), match_reason, flags[].
Company: name, careers_url, city, region, takes_interns (true, false,
unknown), evidence_url, contacts[] (name, role, email or null, source_url),
pitch_draft.
Application: listing_or_company_id, status (saved, applied, interviewing,
offer, rejected), dates, notes.

# Pipeline
1. Fetch from sources listed in the Data sources table: Greenhouse, Lever and
Ashby public board endpoints for a company slug list in
config/companies.json; community summer internship lists on GitHub; remote
job APIs; Hacker News Who is hiring; my Lagos seed list (config/lagos.json).
2. Keep only internships and early-career roles for summer 2027, or roles
with no date that are clearly intern-level.
3. Normalize to the Listing model, dedupe by source plus external_id and by
company plus title, upsert using content_hash.
4. Call Gemini once per batch of new or changed listings with a JSON schema
response: extract duration, intern type, skills, deadline, eligibility,
flags, then score against my profile with a one-sentence reason. Scam flags:
pay-to-apply, no verifiable company, Telegram or WhatsApp-only contact.
5. Mark listings not seen for 14 days as closed. Write data files and a
data/meta.json with the last run time and per-source counts and errors.
6. A failing source must never fail the run: log it in meta.json and
continue.

# Screens
Feed, Listing detail (side pane on desktop, bottom sheet on mobile), Tracker,
Companies (watchlist, contacts, pitch drafter), Settings (profile, FAQ, data
sources, theme, export and import).

# Pitch drafter (on demand, server route)
Input: company page text I provide or the pipeline fetched, plus my profile.
Output under 120 words: one specific line about the company taken from its
text, one project link with UTM parameters, one clear ask. If the text gives
nothing specific, say so and ask me for a line. Never invent facts.
```

```
# UI
Follow the Design system and UI rules sections exactly: luxury typography,
one accent, no gradients, real Lucide icons, one corner radius token, flat
layout with hairlines, no looping animations, and every item marked Yes or
Adapted. Do not build items marked Skip.

# Done means
Each phase meets its acceptance check, the app builds with no type errors,
Lighthouse accessibility is 95 or higher, and I can run everything on free
tiers with a README listing the environment variables and how to add a
company.
```

Build phases

Build in six phases, riskiest data work first, and do not start a phase until the previous acceptance check passes.

1. **Source check.** Pipeline skeleton fetches from every source in the Data sources table. Accept: data/meta.json lists each source with a count or an error, at least three sources return real listings, and nothing from LinkedIn, Indeed or Glassdoor is touched.
2. **Normalize, dedupe, score.** Listing model, dedupe, Gemini extraction and match scoring, eligibility and scam flags. Accept: at least 50 listings with every field filled or null, each with a source URL, and a second run makes zero Gemini calls for unchanged listings.
3. **Feed and detail.** Feed, filters (five visible), detail pane and sheet, skeletons, stale-while-revalidate, command menu, dark mode, site search. Accept: cached feed shows in under 400ms, Lighthouse accessibility is 95 or higher, and the design checklist (UI rules and cleanup rules) passes on every screen.
4. **Tracker and watchlist.** IndexedDB state, optimistic updates, undo toasts, deadline countdowns, goal progress, export and import. Accept: save, dismiss and status changes survive a reload and undo works within 6 seconds.
5. **Contacts and pitches.** Company cards, public contact extraction, LinkedIn search links, pitch drafter behind the passphrase gate. Accept: every contact shows its source page, no email appears without one, and drafts stay under 120 words with no invented facts.

6. **Polish.** View transitions, swipe-to-dismiss sheets, print stylesheet, shortcut hints, scroll progress, weekly digest, optional Supabase sync. Accept: all items marked Yes in the UI rules work, and nothing animates in a loop.